

Online Resource 1

Article: Watermarking Turkish LLM Output: Detector Calibration, Scheme Fragility, and a Released Evaluation Benchmark

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Table S1

Table S1. Realized false-positive rate of the clean-calibrated threshold on attacked negatives, for all 33 scheme-condition cells. The table is descriptive. The 95% upper bound is a binomial bound over 96 rows, but those rows are four per prompt across 24 prompt clusters, so the bound is anticonservative, and the threshold is estimated from the same small clean sample. The threshold is set to 1% FPR on the 96 clean negatives. The fourth column is the rate that same threshold actually achieves once the negatives have been through the attack. The last column gives AUROC when both classes carry the same transformation, which is an ecologically meaningful question for diacritic stripping and translation but not for the laundering attacks, since nobody launders human text to remove a watermark it does not carry.

Scheme	Condition	TPR	Realized FPR	Exceeding	FPR 95% upper	Same-transform AUROC
EXP	clean	1.000	0.010	1/96	0.057	1.000
EXP	dia100	0.917	0.042	4/96	0.103	0.982
EXP	dia50	1.000	0.031	3/96	0.089	1.000
EXP	morph	1.000	0.010	1/96	0.057	1.000
EXP	morph+dia	0.875	0.042	4/96	0.103	0.981
EXP	morph_v1	1.000	0.010	1/96	0.057	1.000
EXP	morph_v1+dia	0.917	0.021	2/96	0.073	0.980
EXP	rtt	0.792	0.021	2/96	0.073	0.960
EXP	para	1.000	0.010	1/96	0.057	1.000
EXP	launder	0.990	0.031	3/96	0.089	0.996
EXP	launder_ap i	0.490	0.062	6/96	0.131	0.840
KGW	clean	1.000	0.010	1/96	0.057	1.000
KGW	dia100	0.865	0.010	1/96	0.057	0.993
KGW	dia50	0.979	0.000	0/96	0.038	0.999
KGW	morph	1.000	0.010	1/96	0.057	1.000

Scheme	Condition	TPR	Realized FPR	Exceeding	FPR 95% upper	Same- transform AUROC
KGW	morph+dia	0.865	0.010	1/96	0.057	0.994
KGW	morph_v1	1.000	0.021	2/96	0.073	1.000
KGW	morph_v1 +dia	0.844	0.010	1/96	0.057	0.995
KGW	rtt	0.594	0.010	1/96	0.057	0.961
KGW	para	0.990	0.000	0/96	0.038	0.999
KGW	launder	0.938	0.000	0/96	0.038	0.999
KGW	launder_ap i	0.427	0.000	0/96	0.038	0.941
SynthID	clean	1.000	0.010	1/96	0.057	1.000
SynthID	dia100	0.677	0.021	2/96	0.073	0.927
SynthID	dia50	0.948	0.010	1/96	0.057	0.996
SynthID	morph	1.000	0.021	2/96	0.073	1.000
SynthID	morph+dia	0.656	0.021	2/96	0.073	0.924
SynthID	morph_v1	1.000	0.031	3/96	0.089	1.000
SynthID	morph_v1 +dia	0.656	0.062	6/96	0.131	0.915
SynthID	rtt	0.312	0.042	4/96	0.103	0.825
SynthID	para	0.990	0.021	2/96	0.073	0.998
SynthID	launder	0.948	0.021	2/96	0.073	0.980
SynthID	launder_ap i	0.250	0.010	1/96	0.057	0.807

Source: results/detection_metrics.csv in the released repository; regenerated by pilot/make_paper_numbers.py.